12TH CHEMISTRY · CHAPTER 21 · CHAPTER TEST

# Halogenoalkanes

Chapter-wise test: 6 MCQs, 5 of 6 short questions, 1 long question

| | | |
|---|---|---|
| **Class:** 12th (HSSC Part II) | **Subject:** Chemistry | **Total Marks:** 26 |
| **Chapter:** 21 | **Time Allowed:** 45 minutes | **Name:** _____ |

## OBJECTIVE PART (6 × 1 = 6)

**Q.1 Choose the correct answer.**

**1. Which is a tertiary halogenoalkane?**

(a) Chloroethane (b) 2-Chloropropane
(c) 1-Chloropropane (d) 2-Chloro-2-methylpropane

**2. The most reactive halogenoalkane is:**

(a) Fluoroalkane (b) Iodoalkane
(c) Bromoalkane (d) Chloroalkane

**3. Primary halogenoalkanes undergo substitution by:**

(a) SN2 (b) SN1
(c) E1 (d) Free radical

**4. Bromoethane with ethanolic KCN gives:**

(a) Ethanenitrile (b) Ethanamine
(c) Propanenitrile (d) Ethene

**5. The precipitate of silver chloride is:**

(a) Cream yellow (b) White
(c) Bright yellow (d) Black

**6. The rate of an E1 reaction depends on the concentration of:**

(a) Base only (b) Both
(c) Substrate only (d) Neither

## SUBJECTIVE PART — SHORT QUESTIONS (5 × 2 = 10)

**Q.2 Write short answers to any FIVE (5) questions.**

i. Why do fluoroalkanes rarely undergo substitution?
ii. Why is the thionyl chloride method useful?
iii. Give the order of stability of carbocations.
iv. What is a beta-elimination reaction?
v. Why does iodoethane give a precipitate with silver nitrate earlier?
vi. Why is chlorobenzene less reactive than chloroethane?

## LONG QUESTION (2 × 5 = 10)

**Q.3 Answer both parts.**

(a) Describe the SN1 and SN2 mechanisms of halogenoalkanes.
(b) Describe four methods of preparation of halogenoalkanes.

studysapiens.com